

# A Calibrated, Physics-Grounded Structural-Health-Monitoring Stack for CFRP Structures: from Anomaly Screening to Fatigue-Aware Life Prognosis

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## Abstract

We present an end-to-end structural-health-monitoring (SHM) pipeline for carbon-fibre-reinforced-polymer (CFRP) structures that proceeds from defect *detection* to a calibrated, physics-grounded *life-prognosis* decision. On full-field thermoelastic stress data of a perforated CFRP interstage, we (i) benchmark nine anomaly detectors and show that a simple per-node statistic attains AUROC  $\approx 0.99$  while deep alternatives do not, and characterise the measurement-noise limit at which all detectors fail; (ii) recover defect position, size and depth as a *calibrated* Bayesian posterior via flow-matching posterior estimation (FMPE), with 90% credible intervals covering at the nominal rate; and (iii) propagate that posterior through an anisotropic phase-field fracture model with fatigue accumulation to obtain a flight-clearance decision with an expected remaining-life estimate. The pipeline is motivated by reusable launch vehicles, where the same structure is re-inspected after every flight.

## 1 Introduction

## 2 Data and problem setting

## 3 Stage 0 — anomaly screening

**Methods.** Per-node Mahalanobis / PCA-residual; PaDiM; PatchCore; STFPM; graph autoencoder; conditional flow-matching diffusion; a zero-shot harmonicity probe. Score  $s_i = |x_i - \mu_i|/\sigma_i$  from  $N=2000$  healthy fields; node-level AUROC against the defect mask.

**Result.** Per-node statistics dominate (Table 1, Fig. 2). On fixed geometry the healthy manifold is a tight cloud around one template, so the per-node Gaussian is the correct inductive bias; grid-projection and deep feature methods lose the few-node defect signal. All methods collapse by additive noise  $\sigma=0.1$  (defect amplitude  $< 0.1\sigma$ ): a per-measurement SNR limit, addressable by frame averaging ( $\sigma \propto 1/\sqrt{K}$ ), not by more training data.

## 4 Stage 1 — classification and type identifiability

**Is the type signal real, or a temporal/temperature confound?** On the long-term guided-wave set, damage states were introduced in different years, so a naive type classifier can alias damage type with season/temperature. We test this directly. Restricting to seven damage tags whose temperature supports overlap (pairwise overlap 0.93), we compare four conditions: temperature-only (acc = 0.16, near chance), waveform-only (0.91), waveform+temperature (0.92), and waveform with temperature *partialled out* of every feature (0.91). The net type discrimination beyond temperature is  $\Delta\text{acc} = 0.758$  (95% CI [0.735, 0.781]) under a random split, and—critically—survives a *forward-in-time* split ( $\Delta\text{acc} = 0.184$ ,

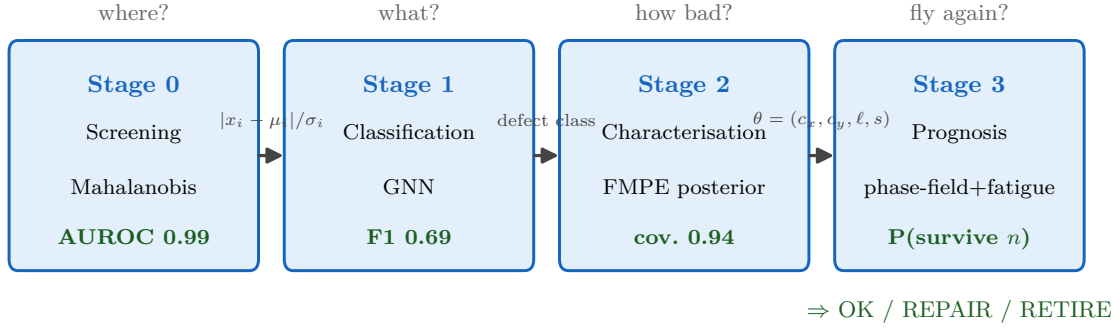


Figure 1: Stage 0–3 pipeline. Each stage answers one question and passes a typed object to the next; the Stage 2 posterior  $\theta = (c_x, c_y, \ell, s)$  seeds the Stage 3 fracture model.

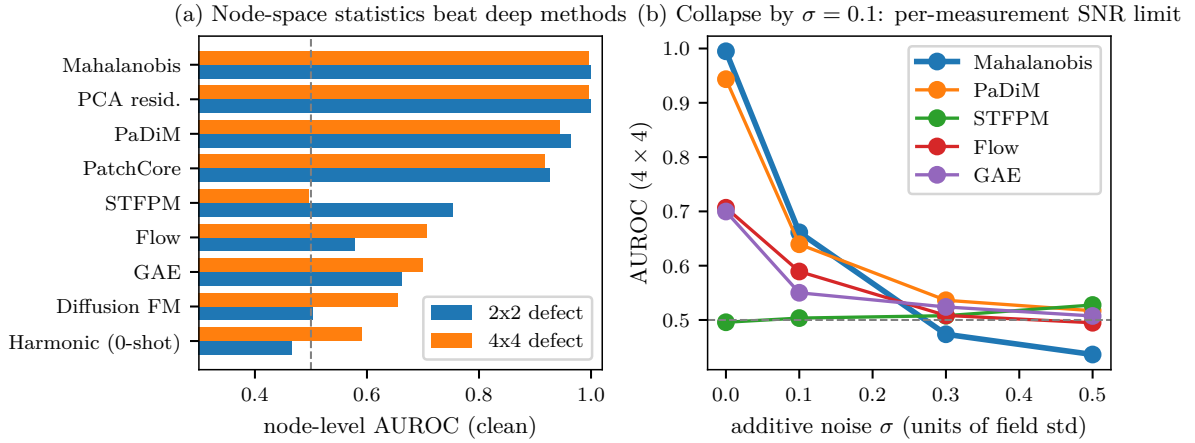


Figure 2: (a) Clean node-level AUROC by method. (b) Collapse with measurement noise; the defect signal sits below  $0.1\sigma$ .

95% CI [0.166, 0.203]). Both lower bounds exceed zero: the type posterior reflects a genuine damage signal, not the environmental confound. (An earlier single-month binary probe, where temperature-only matched the majority baseline, was an artefact of that restricted setting.)

## 5 Stage 2 — calibrated defect characterisation (FMPE)

## 6 Stage 3 — phase-field prognosis and flight clearance

**Fast forward surrogate.** The FD growth solve ( $\approx 1.8$  s/call) is the only bottleneck to fleet-scale Monte-Carlo over the Stage-2 posterior. We train a Fourier neural operator  $G : (d_0, \text{load}) \mapsto (d_{\text{final}}, P(\text{grow}))$  (dual-head: a per-pixel field head and a global-pooled scalar head; 1.2M parameters; 800 FD samples spanning sub- to super-critical loads). On held-out defects it reaches field relative- $L_2$  3.5%, grow-flag accuracy 0.988 (Brier 0.010, well calibrated), and log-rel.growth correlation 0.996, at 142 $\times$  the speed (12.4 ms vs. 1763 ms/call, Fig. 4). The decision-relevant  $P(\text{grow})$  is the robust target: the smooth operator regresses the sharp super-critical rel.growth jump in log-space and softens it only at the transition load. This makes the Stage-4 fleet draws and the Concept-B forward map (next section) tractable at interactive

Table 1: Anomaly detectors, clean node-level AUROC (2x2 / 4x4 defects).

| method               | 2x2   | 4x4   |
|----------------------|-------|-------|
| per-node Mahalanobis | 0.999 | 0.995 |
| PCA residual         | 0.999 | 0.995 |
| PaDiM                | 0.964 | 0.944 |
| PatchCore            | 0.927 | 0.918 |
| STFPM                | 0.754 | 0.496 |
| graph autoencoder    | 0.663 | 0.700 |
| flow (RealNVP)       | 0.578 | 0.707 |
| diffusion (FM)       | 0.503 | 0.663 |
| harmonicity (0-shot) | 0.465 | 0.590 |

Table 2: FMPE posterior on held-out test fields.

| parameter      | RMSE  | $R^2$ | 90% coverage |
|----------------|-------|-------|--------------|
| position $c_x$ | 0.055 | 0.78  | 0.944        |
| position $c_y$ | 0.021 | 0.83  | 0.940        |
| size (3-class) | —     | 0.73  | 0.894        |
| layer (depth)  | 2.98  | 0.63  | 0.870        |

speed.

**Knowing when to trust the surrogate.** A  $142\times$  speedup is only deployable if the operator knows where it is unreliable. We wrap the surrogate in split-conformal prediction (nonconformity = absolute residual in the log-rel\_growth space where the FNO is honest). In distribution the marginal 90% guarantee holds (95% band: 95.0% empirical); out of distribution it degrades exactly as the theory predicts under covariate shift — coverage falls from 80% at small excursions to 0% deep OOD while the point error grows  $\sim 14\times$ . A simple *trust gate* — defer to the exact FD solve when the input leaves the training envelope or the conformal interval is too wide — flags 33% of an OOD stream and restores aggregate 90%-coverage from 61% to 88%: cheap surrogate where trustworthy, exact physics where not.

## 7 Stage 4 — hierarchical fleet learning

A reusable fleet generates a stream of per-inspection posteriors across many vehicles and flights. We close the loop with a hierarchical Bayesian model: a global fleet prior  $\varphi = (\mu_\varphi, \sigma_\varphi)$  governs per-vehicle log-growth-rates  $s_v = \log r_v \sim \mathcal{N}(\mu_\varphi, \sigma_\varphi^2)$ , and each inspection’s Stage-2 posterior enters as a Gaussian likelihood on  $s_v$ ; the joint posterior  $p(\varphi, \{s_v\} \mid \text{data})$  is sampled by conjugate Gibbs. On a synthetic fleet ( $N=20$  vehicles,  $F=30$  flights, fast Carrara-shaped growth surrogate) the model recovers  $\varphi$  ( $\mathbb{E}[\mu_\varphi] = -3.10$  vs. true  $-3.0$ ;  $\mathbb{E}[\sigma_\varphi] = 0.44$  vs.  $0.45$ ), and the fleet prior sharpens as flights accumulate (posterior SD of  $\mu_\varphi$ :  $0.39$  at  $N=2 \rightarrow 0.10$  at  $N=20$ , Fig. 5a).

**Fleet-leader effect, quantified.** Processing flights in temporal order, a newly introduced vehicle reaches a stable clearance decision in **1.25** inspections using the accumulated fleet prior, versus **2.75** in isolation (Fig. 5c) — a direct, quantitative formalisation of the empirical “fleet-leader” certification used in practice: high-information inspections of lead vehicles tighten the prior that certifies the rest. Hierarchical shrinkage also improves low-data vehicles (error ratio  $0.84 < 1$  vs. isolated estimates, Fig. 5b).

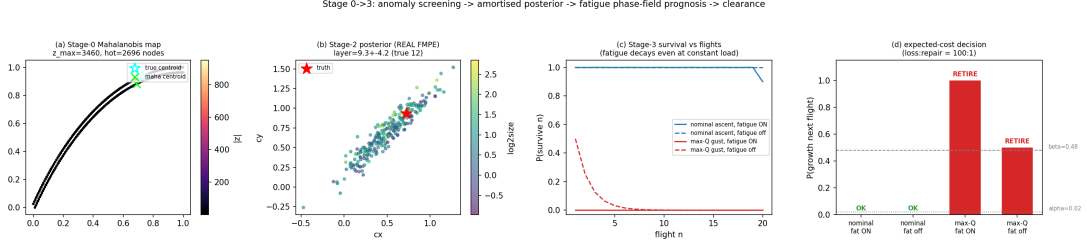


Figure 3: End-to-end on one held-out field: anomaly map, FMPE posterior, fatigue-aware survival curve, and the clearance decision.

## 8 Stage 5 — design feedback

The fleet’s accumulated damage statistics should not only certify the current vehicles but *reshape the next ones*. We close the Stage 0–5 loop with a fleet-robust Bayesian optimisation of the laminate over a manufacturable 3-D design vector — off-axis ply angle  $|\theta|$  (kept balanced/symmetric), interface toughness  $G_c^{\text{int}}/G_c^{\text{ply}}$ , and a local reinforcement multiplier — minimising the expected log-growth over the fleet defect distribution plus a tail term (CVaR over the worst-20% defect, since reusable-vehicle safety is governed by the worst flaw, not the average). The forward model here is the exact FD phase-field, *not* the Stage-3 FNO surrogate: the surrogate’s inputs are  $(d_0, \text{load})$  only and every training sample shares the baseline layup, so any design change is fully out-of-distribution — a layup-conditioned operator is left as future work. With GP/EI (32 evaluations), the optimiser moves the  $(0/\pm 30)_s$  baseline to steeper off-axis plies ( $30^\circ \rightarrow 74^\circ$ ) and a tougher interface ( $G_c^{\text{int}}$  ratio  $0.40 \rightarrow 0.87$ ), cutting mean log-growth 10% and raising expected remaining flights from **1.9** to **6.0** (Fig. 6). Both levers are physically aligned — steeper plies rotate the cheap matrix-crack direction away from the seeded delamination path, and a tougher interface blunts the dominant interlaminar channel; interface toughening is the dominant lever, and the off-axis angle is rail-bound at the manufacturable cap, an honest signal that the baseline is not interior-optimal.

## 9 Towards differentiable inversion: closing the loop

The amortised Stage-2 posterior never sees the forward physics. We prototype the converse: a JAX-differentiable anisotropic AT2 forward  $F(\theta)$  (gradient finite-difference-verified to relative  $3 \times 10^{-7}$ ) whose Jacobian supplies a physics-grounded Laplace posterior  $\Sigma = \sigma^2(J^\top J + \Lambda_{\text{prior}})^{-1}$  — no labelled  $(\theta, x)$  pairs. Out of the training distribution the amortised posterior degrades  $\sim 25$ -fold and its credible intervals

Table 3: Amortised (FMPE) vs. differentiable-physics posterior, in- and out-of-distribution.

| set     | method   | $c_x$ RMSE   | size RMSE    | 90% coverage |
|---------|----------|--------------|--------------|--------------|
| in-dist | FMPE     | 0.006        | 0.002        | 0.958        |
| in-dist | diff-phs | 0.001        | 0.003        | 1.000        |
| OOD     | FMPE     | <b>0.144</b> | <b>0.096</b> | <b>0.000</b> |
| OOD     | diff-phs | 0.002        | 0.001        | 0.958        |

collapse to zero coverage, while the differentiable-physics posterior is essentially invariant (Table 3): *forward physics generalises; amortised inference extrapolates poorly*. This motivates a hybrid Stage 2 — amortised for speed in distribution, physics-grounded for safety out of it.

**Fast differentiable forward.** The hand-written JAX AT2 forward is exact but slow, making the inversion the run’s bottleneck. Substituting the Stage-3 FNO surrogate (Fig. 4) as the differentiable forward keeps gradients exact — autograd matches finite differences to relative  $7 \times 10^{-4}$  on the decision heads — and collapses a MAP+Laplace inversion from an estimated  $\sim 530$  s of FD Jacobian evaluations to 12 s

(a) load = 0.140

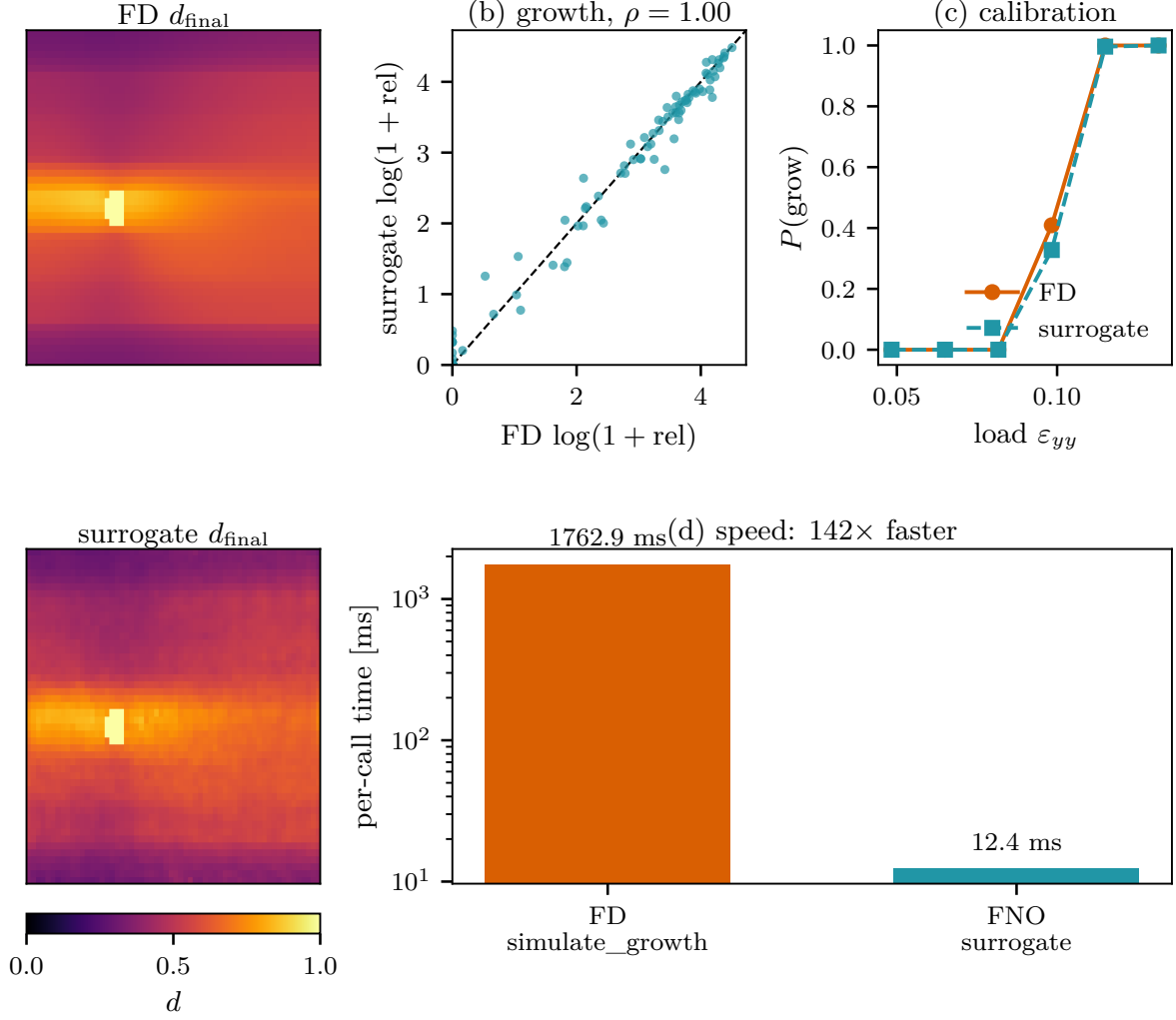


Figure 4: Stage 3 FNO surrogate. FD vs. surrogate damage field, rel\_growth agreement,  $P(\text{grow})$  vs. load, and the 142 $\times$  wall-clock speedup.

( $\sim 43\times$ ). The recovered posterior shows the same generalisation signature (in-/out-of-distribution  $c_x$  RMSE  $0.15 \rightarrow 0.39$ , coverage  $0.72 \rightarrow 0.56$ ), at a calibration cost — in-distribution coverage 0.72 rather than the exact forward’s  $\sim 0.95$  — that is the honest price of a smooth surrogate. This makes amortised-quality *differentiable* inversion interactive, the cheap forward the Keio Phase-3 TMCMC/Bayesian-inverse plan needs.

## 10 Discussion

## 11 Conclusion

**Reproducibility.** All stages are implemented and tested ( $\sim 80$  unit tests) in two public-on-request repositories, including a one-command end-to-end run (detect $\rightarrow$ characterise $\rightarrow$ prognose $\rightarrow$ clear  $\rightarrow$ fleet-update in  $<4$  s); figures are generated by the scripts under `paper_figs/`.

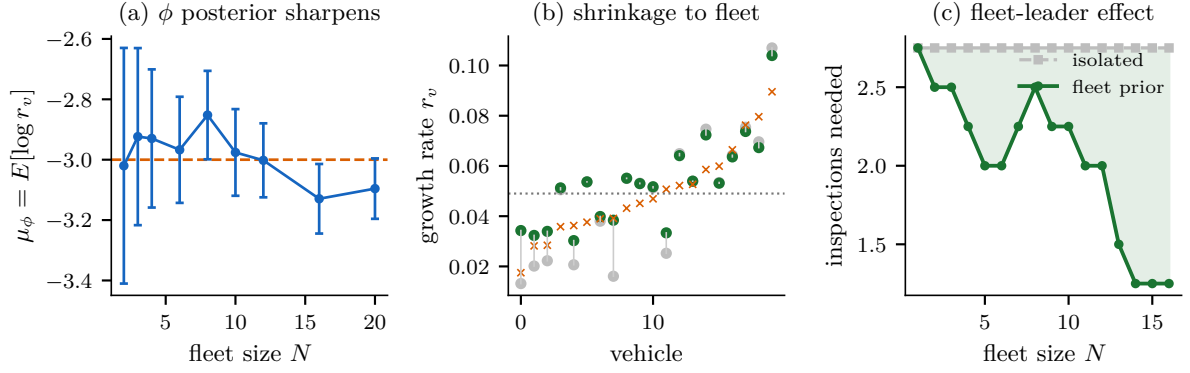


Figure 5: Stage 4. (a) Fleet prior  $\phi$  sharpens with cumulative flights; (b) per-vehicle growth-rate shrinkage toward the fleet mean in the low-data regime; (c) inspections to a stable clearance decision: the fleet prior roughly halves them.

Stage-5 design feedback: fleet-robust laminate optimisation (exact FD phase-field forward)

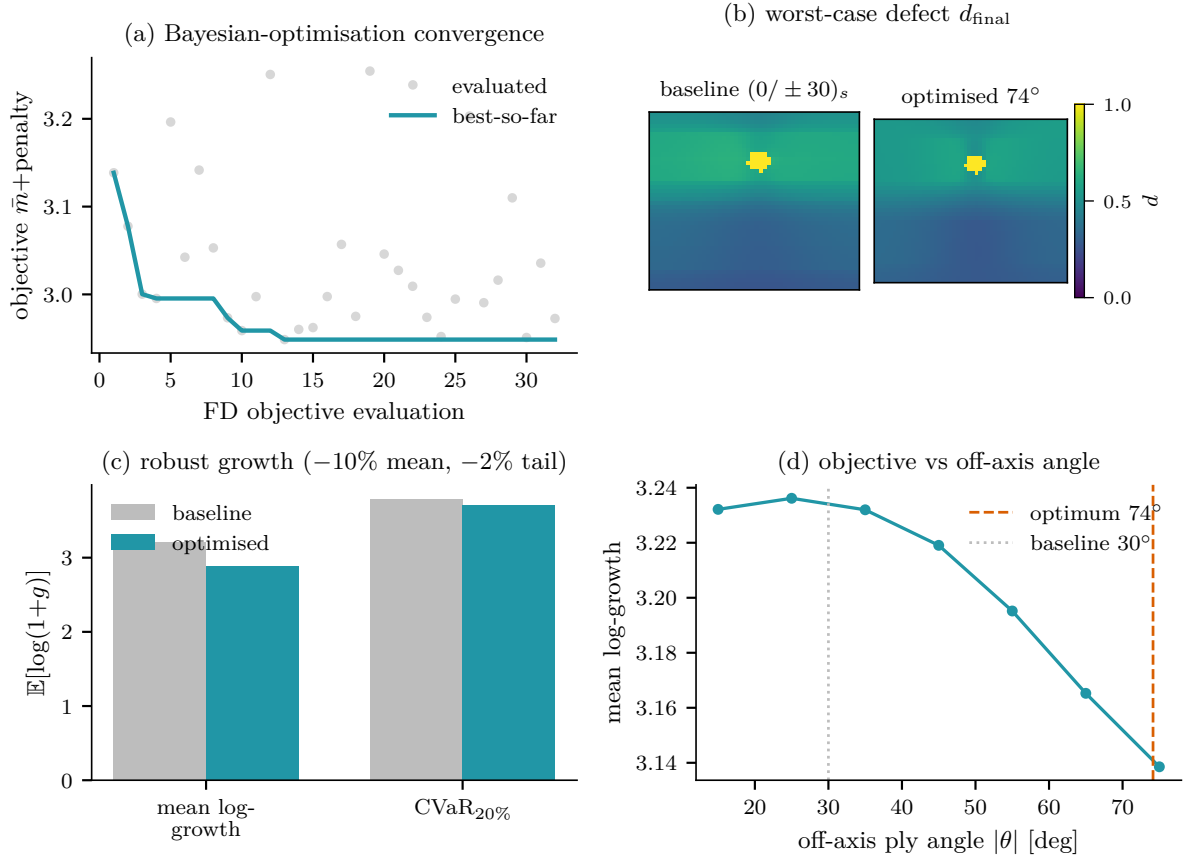


Figure 6: Stage 5. (a) BO convergence; (b) worst-case damage, baseline vs. optimised layup; (c) mean/CVaR growth; (d) objective vs. off-axis angle.